MILITARY SPECIFICATION

GENERATOR, PULSE: 10526464

1. SCOPE

1.1 This specification covers one type of pulse generator, part of the Gun Direction Computer: M18, used to generate synchronizing pulses for triggering of the logic circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18, Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10526464	Generator, Pulse

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

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(Copies of specifications, standards, drawings and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The pulse generator shall be manufactured in accordance with Drawing F10526464 and drawings pertaining thereto.

3.2 General specifications.- The pulse generator shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357.

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The pulse generator shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the pulse generator for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the pulse generator shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The pulse generator shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The pulse generator shall meet the requirements of 3.4.2.2 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The pulse generator shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The pulse generator shall meet the requirements of 3.4.2.2 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating.- The pulse generator shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) and atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The pulse generator shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The pulse generator shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 0.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the pulse generator assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%. The requirements of 3.4.1 thru 3.4.2.1 shall apply with the following prevalent conditions:

a. The following voltages shall be applied as specified:

(1) -18 volts, direct current (vdc) to pin 62A
(2) -10 vdc to pins 33B and 34B
(3) -6 vdc to pins 25A and 26A
(4) -3 vdc to pins 59A and 39B
(5) +1.25 vdc to pin 44A
(6) +3 vdc to pins 46B, 28A, 29A, 30A and 31A
(7) +6 vdc to pins 30B and 31B
(8) +15 vdc to pin 32B

b. The dc returns of aforementioned voltage levels shall be connected to the following pins which have been shorted together: 1A, 2A, 3A, 4A, 5B, 6B, 7B, 8B, 10A, 11A, 12A, 13A, 14A, 15A, 17A, 18A, 19A, 20A, 38A, 39A, 40A, 43B, 49B, 50B and 51B.

3.4.1 Static performance.

3.4.1.1 Transistor Q1, shall have a dc beta of greater than 30.

3.4.1.2 Transistors Q4, Q6, Q8, Q11 shall be biased for linear, class A, operation.

3.4.1.3 Transistors Q14 and Q16 shall be biased off under static conditions.

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3.4.1.4 With the conditions specified in paragraph 3.4 and a -6Vdc voltage applied to pins 41B and 45A the following requirements shall be met:

a. When Q13 is OFF, Q14 shall be ON and Q16 shall be OFF.
b. When Q13 is ON, Q14 shall be OFF and Q16 shall be ON.
c. The forward voltage drop across CR19 plus the forward drop, base-to-emitter, Q14 shall be less than 1.4Vdc.

3.4.1.5 With the conditions specified in paragraph 3.4 and pins 58B and 51B shorted, the following requirements shall be met:

a. The forward voltage drop across CR30 shall be 0.4 ± 0.15Vdc.
b. The forward voltage drop across CR26 and CR29 shall be 0.8 ± 0.2 Vdc.
c. The forward voltage drop across CR25, CR26 and CR29 shall be 1.2 ± 0.3Vdc.
d. The back bias, base to emitter, of transistor Q18 shall be at least 0.4Vdc under static conditions.
e. Transistors Q18 and Q20 are both OFF under static conditions.
f. The forward voltage drop across CR27 plus the forward voltage drop on Q23 shall be less than 1.2Vdc.
g. Transistor Q23 shall be ON. The saturation voltage from collector to emitter shall be less than 0.3Vdc.
h. Transistors Q24 and Q25 shall have a back bias of at least 0.2Vdc from base to emitter.
i. The currents through transistors Q17, Q19, Q21, Q22 shall be equalized. The current through any one of these transistors shall not be greater than 200 milliamperes (ma) nor less than 100 ma.
j. The saturation voltage of transistors Q17, Q19, Q21 and Q23 shall not be greater than 0.5Vdc.
k. The forward voltage drop across diodes CR21, CR23, CR24, CR28 shall be 0.75 ± 0.2Vdc.
l. Transistors Q2, Q3, Q5, Q7, Q9, Q10, Q12 and Q15 shall be biased OFF, under static conditions, by at least 0.3Vdc.
m. The clamp diodes clamp the output voltage at minus 6.3 ± 0.8Vdc, under static conditions.
n. The leakage current in diode CR27 shall be less than 0.5 nanoampere.

3.4.1.6 With the following exception and additions the prevalent conditions specified in paragraph 3.4 shall prevail for requirements 3.4.1.6.1 thru 3.4.1.6.7.

a. The -3Vdc voltage shall be removed.
b. A +15Vdc voltage shall be applied to pin 39B thru a 22 thousand (K) ohm resistive load.
c. The loads shown in Figure 1 and Figure 2 shall be applied to the pins designated in each figure.

3.4.1.6.1 Transistor Q20 shall be ON under unstable conditions. When Q20 is ON its saturation voltage is less than 0.3Vdc.

3.4.1.6.2 Transistor Q23 shall be OFF under unstable conditions. When Q23 is OFF its collector-to-emitter voltage is minus 6.2 ± 1.0Vdc.

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3.4.1.6.3 Transistors Q24 and Q25 shall be ON under unstable conditions. The voltage drop collector to emitter plus the voltage drop across the emitter resistors shall be less than 0.7Vdc.

3.4.1.6.4 Transistors Q17, Q19, Q21, Q22 shall be biased OFF by at least 1Vdc, base-to-emitter in the unstable state.

3.4.1.6.5 The "0" output voltage of the pulse generator shall be more positive than -.85Vdc.

3.4.1.6.6 The "1" output voltage of the pulse generator shall be more positive than -.85Vdc.

3.4.1.6.7 Transistors Q2, Q3, Q5, Q7, Q9, Q10, Q12 and Q15 shall have a current of 220 ± 55 ma in their emitter circuits when the pulse generator is under a static load in the unstable state.

3.4.2 Dynamic performance.

3.4.2.1 With the conditions specified in paragraph 3.4 and a 0.20 ± 0.06V, peak-to-peak sine wave signal at a frequency between 448KC and 493KC, introduced (pins 59B to 50B), the output waveforms at the points listed in Column A, Table I shall be as specified in Column B, Table I. Pins 58B and 51B shall be shorted during the dynamic operation and performance of Q1 and Q4.

TABLE I

A	B
TP1	Figure 3
TP5	Figure 4
Pin 56B	Figure 5
Pin 55A	Figure 6
Pin 48B	Figure 7
TP21	Figure 8
Pin 35A	Figure 9
Pin 4B	Figure 10
Pin 23A	Figure 10
Pin 14B	Figure 11
Pin 6A	Figure 11

3.4.2.2 Circuit outputs (Extreme temperatures).- With the conditions specified in paragraph 3.4 and a 0.20 ± 0.06V, peak-to-peak sine wave signal at a frequency between 448KC and 493KC, introduced (pins 59B to 50B) the circuit outputs of the pulse generator shall be as exhibited in Figure 11.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II, III, IV and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table II shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE II

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. Static performance	3.4.1	4.5.2.1
102. Dynamic performance	3.4.2	4.5.2.2

4.3 Control tests.

4.3.1 General.- One pulse generator shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

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4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table IV.

TABLE IV

CONTROL TEST	REQUIREMENTS	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1, 3.4.2.2	4.5.1.3.2, 4.5.2.2.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.2.2	4.5.1.5, 4.5.2.2.2
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.2).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item	Description
1. Voltage measuring device, dc	Accuracy ± 5 percent of reading, input resistance 100K ohms or greater.

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Item	Description
2. Voltage measuring device, ac	Accuracy ± 1 percent of reading, range 0.1 millivolts to 300 volts. Input impedance 10M ohms or greater and 15 pico-farads or less. Frequency range from 10cps to 5mc or greater.
3. Oscillescope	Accuracy ± 4 percent, rise time 50 nanoseconds or less, sweep period from 50 nanoseconds to 1 millisecond or greater. Input impedance 1M ohm or greater, and 20 pico-farads or less.
4. Attenuating probe	Attenuation factor of 10. Input impedance 10M ohms or greater than 10 pico-farads or less. (Probe to be used with item 3.)
5. Sine wave generator	Frequency 448 kilocycles to 493 kilocycles, output adjustable from 0 to 1 volt or greater, peak-to-peak. Distortion less than 3 percent. Output impedance less than 700 ohms.
6. Power supply	Minus 18 volts; accuracy ± 2 percent, current rating 150 ma or greater, ripple equal to or less than 600 mv peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
7. Power supply	Minus 10 volts; accuracy ± 2 percent, current rating 150 ma or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
8. Power supply	Minus 6 volts; accuracy ± 2 percent, current rating 600 ma or greater, ripple equal to or less than 400 mv peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.

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Item		Description
9.	Power supply	Minus 3 volts; accuracy ± 2 percent, current rating 10 ma or greater, ripple equal to or less than 100 mv, peak-to-peak output impedance equal to or less than 2 ohms from dc to 500 kc.
10.	Power supply	Plus 1.25 volts; accuracy ± 2 percent current rating 20 ma or greater, ripple equal to or less than 100 mv, peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
11.	Power supply	Plus 3 volts; accuracy ± 2 percent, current rating 600 ma or greater, ripple equal to or less than 100 mv, peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
12.	Power supply	Plus 6 volts; accuracy ± 2 percent, current rating 60 ma or greater, ripple equal to or less than 100 mv, peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
13.	Power supply	Plus 15 volts; accuracy ± 2 percent, current rating 50 ma or greater, ripple equal to or less than 100 mv, peak-to-peak, output impedance equal to or less than 2 ohms from dc to 500 kc.
14.	Resistor	1k ohm ± 5 percent 1/4 watt.
15.	Resistor	22k ohm ± 5 percent 1/4 watt.
16.	Capacitor	0.1 microfarad, 100 volt nonpolarized.
17.	Resistor, Carbon (2)	50 ohm ± 5% 1 watt.
18.	Resistor, carbon (2)	30 ohm ± 15% 20 watt.
19.	Capacitor (2)	.01 uf, 50 volts.
20.	Capacitor (2)	.022 uuf, 50 volts.

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the pulse generator rigidly mounted in its normal operating position. Subsequent to the vibration rest, the pulse generator shall show no damage or defect resulting from the test when examined by visual and tactile means. The pulse generator shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the pulse generator shall be shocked in the vertically plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the pulse generator in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced in 2 hours, the 6 hour period reduced to 2 hours and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). While exposed to this test condition the pulse generator shall be subjected to the test specified in 4.5.2.2.2 and shall meet the requirements referenced in 3.3.4.1.

4.5.1.4 Salt spray.- The pulse generator shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the pulse generator shall be inspected for any physical damage, then tested in accordance with 4.5.2.1 to 4.5.2.2.1 for compliance with 3.3.6.

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4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the pulse generator shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the pulse generator shall be tested in accordance with 4.5.2.2.2 to determine compliance with the requirements of 3.4.2.2 as specified in 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the pulse generator shall be tested in accordance with 4.5.2.1 to 4.5.2.2.1 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- One of the first 5 pulse generators produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- Conformance with 3.4 shall be determined by connecting the power supplies specified in Column A of Table V to the respective pins of Column B of Table V. Connect the following pins together; 49B, 50B, 51B, 1A, 2A, 3A, 4A, 5B, 6B, 7B, 8B, 10A, 11A, 12A, 13A, 14A, 15A, 17A, 43B, 18A, 19A, 20A, 38A, 39A and 40A. Connect the power supply ground return to these pins.

TABLE V

Column A	Column B
-18 volt power supply	62A
-10 V	33B and 34B
-6 V	25A and 26A
-3 V	59A and 39B
+1.25 V	44A
+3 V	46B, 28A, 29A, 30A, 31A
+6 V	30B and 31B
+15 V	32B

4.5.2.1 Static performance.- The following tests shall be conducted with the pins and power supplies connected as specified in 4.5.2 to determine compliance with 3.4.1.

4.5.2.1.1 Measure the voltage from pin 59B to pin 60B. The voltage shall be less than plus 0.35 volt.

4.5.2.1.2 Measure the dc voltage from TP1 to pin 51B. The voltage shall be more negative than minus 2.0 volts. Dc beta shall be as specified in 3.4.1.1.

4.5.2.1.3 Measure the dc voltage from the pin or test point specified in Column A of Table VI to pin 51B. The voltage measured shall be as specified in the same row of Column B of Table VI to determine compliance with 3.4.1.2.

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TABLE VI

Column A	Column B
TP2	minus 3.9 ± 0.2 vdc
TP3	minus 8.5 ± 1.0 vdc
TP4	minus 3.7 ± 0.2 vdc
TP5	minus 6.0 ± 0.3 vdc
TP6	minus 5.7 ± 0.4 vdc
Pin 56B	minus 13.0 ± 3.0 vdc
Pin 58A	minus 6.0 ± 0.3 vdc
Pin 48A	minus 5.7 ± 0.4 vdc
Pin 55A	minus 13.0 ± 3.0 vdc
Pin 54B	minus 6.0 ± 0.3 vdc
Pin 52B	minus 5.7 ± 0.4 vdc

4.5.2.1.4 Connect the minus 6 volt power supply to pin 41B, and pin 45A. Measure the voltage from pin 44A to pin 41A. The voltage shall be less than 0.9 vdc to determine conformance with 3.4.1.3.

4.5.2.1.5 Make the following connections and measurements to determine conformance with the requirements of 3.4.1.4:

a. Measure the voltage from pin 44A to pin 44B. The voltage shall be less than 1.4 vdc.

b. Connect pin 43A, through a 1K ohm resistor (Item 14, paragraph 4.4.1.1) to pin 50B. Measure the voltage from pin 46B to pin 44B. The voltage shall be less than plus 0.3 vdc.

c. Measure the voltage from pin 43B to pin 37A. The voltage shall be less than 0.3 vdc. Remove the minus 6 volt power supply from pin 41B and pin 45A. Remove the resistor connected between pin 50B and 43A.

4.5.2.1.6 Make the following connections and measurements to determine conformance with 3.4.1.5:

a. Connect pin 58B to pin 51B. Measure the voltage from the pin or test point indicated in Column A of Table VII to the corresponding pin or test point in Column B of Table VII. The voltage is indicated in the corresponding row of Column C.

TABLE VII

Column A pin or test point	Column B pin or test point	Column C volts dc
Pin 36A	Pin 33A	plus 0.4 ± 0.15
Pin 35B	Pin 36A	plus 0.8 ± 0.2
TP13	Pin 36A	plus 1.2 ± 0.3
TP9	Pin 36A	greater than plus 0.4
Pin 36A	TP11	plus 8.8 ± 0.8
Pin 36A	TP16	less than 1.2
Pin 36A	TP21	less than 0.3
TP21	Pin 33A	greater than plus 0.2
Pin 41A	Pin 51B	greater than plus 0.4
Pin 37A	Pin 51B	more negative than minus 9.0

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b. Measure the voltage from pin 28A to each of the following test points: TP8, TP12, TP15, TP20. Each of these four voltages shall be within ± 20 percent of any of the other three.

c. Measure the voltage from each of the test points in Table VIII to TP18. Each voltage shall be less than 0.5 vdc.

TABLE VIII

Test Points

TP8
TP12
TP15
TP20

d. Measure the voltage from TP18 to each of the following test points: TP7, TP10, TP14, TP19. Each of these four voltages shall be plus 0.75 ± 0.2 vdc.

e. Measure the voltage from pin 23A to 1A. The voltage shall be greater than plus 0.3 vdc.

f. Measure the voltage from pin 3B to 1A. The voltage shall be greater than plus 0.3 vdc.

g. Measure the voltage from pin 3A to pin 5A. The voltage shall be plus 6.3 ± 0.8 vdc.

h. Measure the voltage from pin 3A to pin 15B. The voltage shall be plus 6.3 ± 0.8 vdc.

4.5.2.1.7 Make the following connections and measurements to determine conformance with the requirements of 3.4.1.6.

a. Connect the loads shown in Figure 1 and Figure 2 to the pins designated in the respective figures.

b. Remove the minus 3 volt power supply from pin 39B. Connect pin 39B to the plus 15 volt power supply through a 22K resistor (Item 15, paragraph 4.4.1.1).

c. Measure the voltage from the plus 15 volt power supply to pin 39B. The voltage shall be less than 11 millivolts.

d. Measure the voltage from pin 36A to TP11. The voltage shall be less than 0.3 vdc.

e. Measure the voltage from pin 36A to TP21. The voltage shall be plus 6.2 ± 1.0 vdc.

f. Measure the voltage from pin 33A to pin 35A. The voltage shall be less than plus 0.7 vdc.

g. Measure the voltage from pin 35A to pin 28A. The voltage shall be greater than plus 1 vdc.

h. Measure the voltage from pin 1A to pin 5A. The voltage shall be more positive than -0.85 vdc.

i. Measure the voltage from pin 1A to 14B. The voltage shall be more positive than -0.85 vdc.

j. Measure the voltage from pin 1A to each of the pins listed in Table IX. In each case the voltage shall be plus 300 ± 75 millivolts dc.

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TABLE IX

1B
2B
10B
11B
12B
20B
22B
24B

4.5.2.1.8 Upon completion of the static tests, remove the connection from pin 58B to pin 51B. Remove the plus 15 volt power supply and the 22K ohm resistor from pin 39B. Connect the minus 3 volt power supply to pin 39B. This will return the test equipment to the state specified in 4.5.2.1.

4.5.2.2 Dynamic performance.- These following tests shall be conducted with the pins and power supplies connected as specified in 4.5.2 and the loads shown in Figures 1 and 2 connected to the pins designated in the respective figures.

4.5.2.2.1 Make the following connections and measurements to determine conformance with the requirements of 3.4.2.1. Connect the generator specified in paragraph 4.4.1.1 Item 5 to pin 59B through a 0.1 microfarad capacitor (Item 16, paragraph 4.4.1.1. Connect the ground return of the generator to pin 50B and adjust the sine wave generator so that the output amplitude of the sine wave is 0.20 ± 0.06 volt, peak-to-peak. Connect the oscilloscope specified in paragraph 4.4.1.1, Item 3 (using the attenuating probe specified in paragraph 4.4.1.1, Item 4), to the pin or test point indicated in Column A of Table X. The wave form to be observed at each of these points will be as shown in the figure indicated in Column B of Table X. Special conditions for each test are given in Column C of Table X.

TABLE X

Column A	Column B Figure	Column C Special Conditions
TP1	3	Connect Pin 58B to Pin 51B
TP5	4	Connect Pin 58B to Pin 51B
Pin 56B	5	None
Pin 55A	6	None
Pin 48B	7	None
TP21	8	None
Pin 35A	9	None
Pin 4B	10	None
Pin 23A	10	None
Pin 14B	11	None
Pin 6A	11	None

4.5.2.2.2 Circuit output (Extreme temperatures).- This test shall be conducted at the temperatures specified in 3.3.3.1 by using the test set-up specified in 4.5.2.2 and 4.5.2.2.1. Connect the oscilliscope specified in

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4.4.1.1, Item 3 (use the attenuating probe specified in 4.4.1.1, Item 4), to pin 6A and observe the waveform. Remove the test lead from pin 6A and connect it to pin 14B and again observe the waveform. The waveform at pins 6A and 14B shall be as shown in Figure 11 to determine compliance with 3.4.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Packaging, packing and marking shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The pulse generator is intended to form, amplify and generate two output signals to be used for synchronizing the computer circuits. The two outputs basically a primary and secondary, are derived from a synchronous 448 kilocycle sine wave signal, permanently recorded on the timing or clock channel of the memory disk. The primary output is a clock pulse consisting of a 0.5 microsecond false state pulse (0 volt level) and a 1.8 microsecond true state pulse (minus 6 volt level) intended to trigger all logic flip-flops. The secondary output or strobe pulse is an inverted clock pulse intended to gate the read amplifiers.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)
(e) A block of serial numbers to identify each pulse generator

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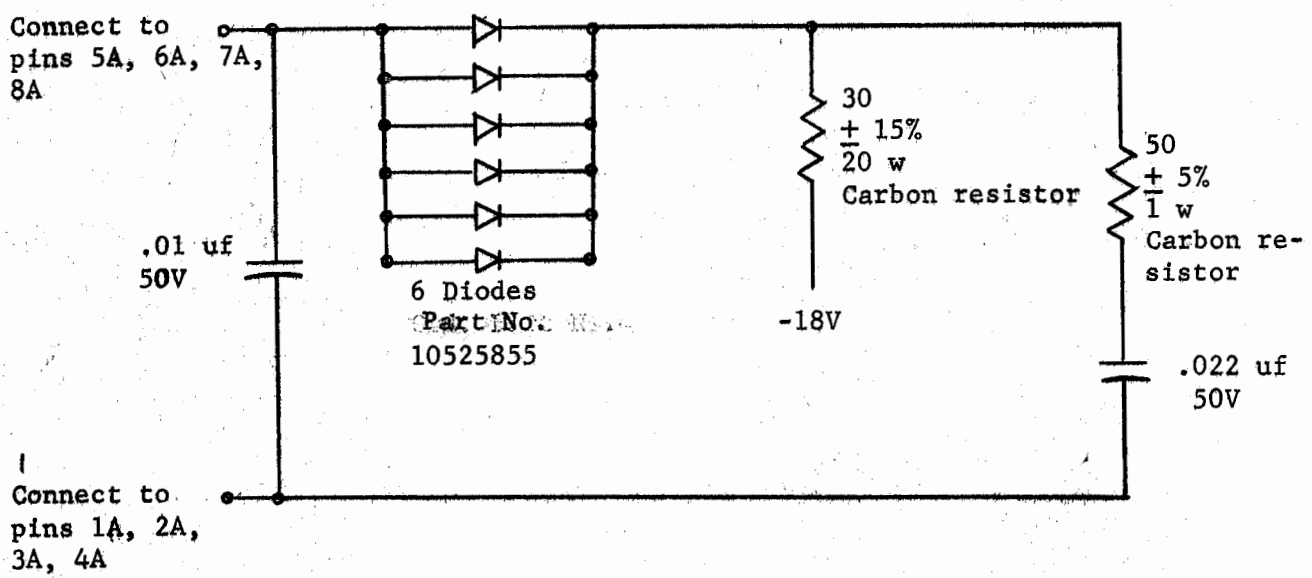


FIGURE I

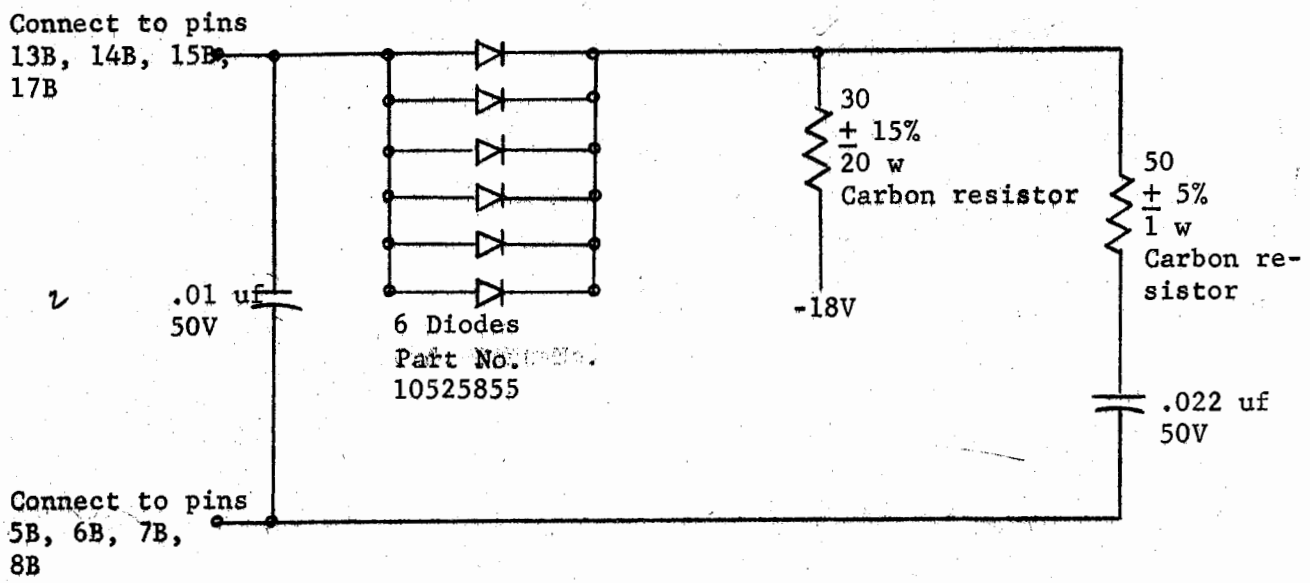
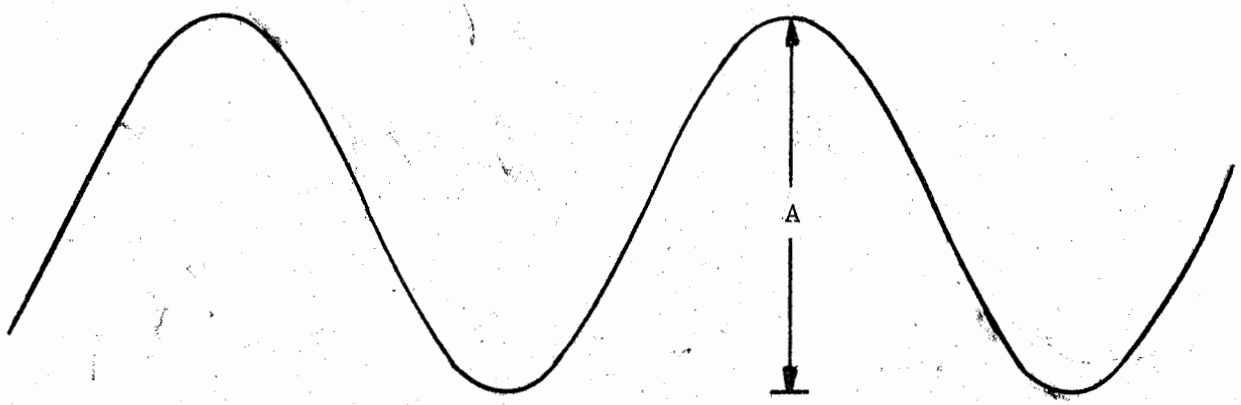


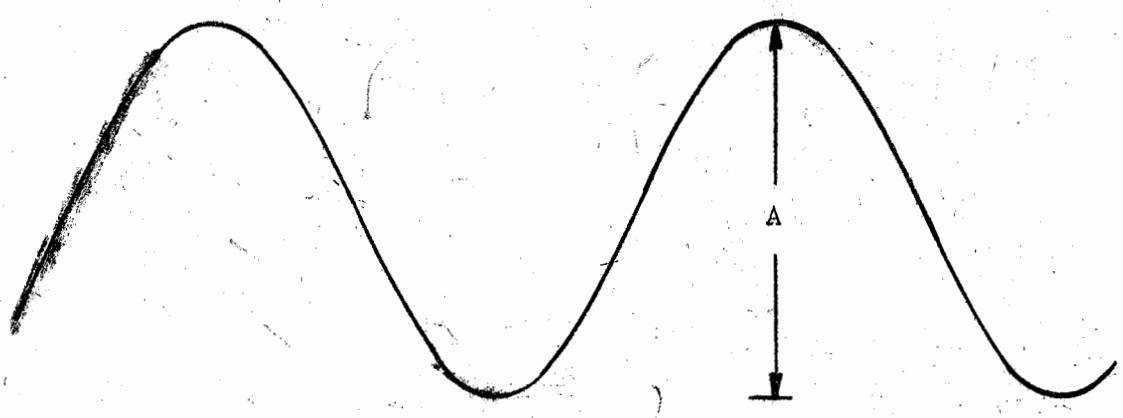
FIGURE 2

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A - Greater than 0.9 times the input signal

FIGURE 3



A - Greater than 4 times the input signal

FIGURE 4

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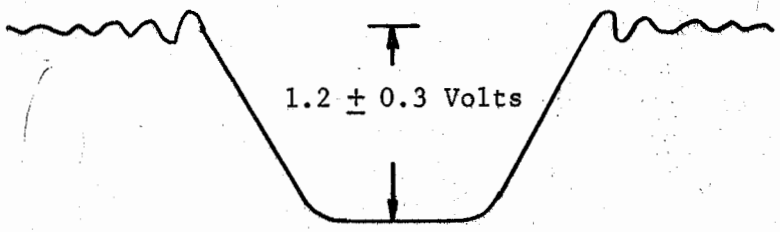


FIGURE 5

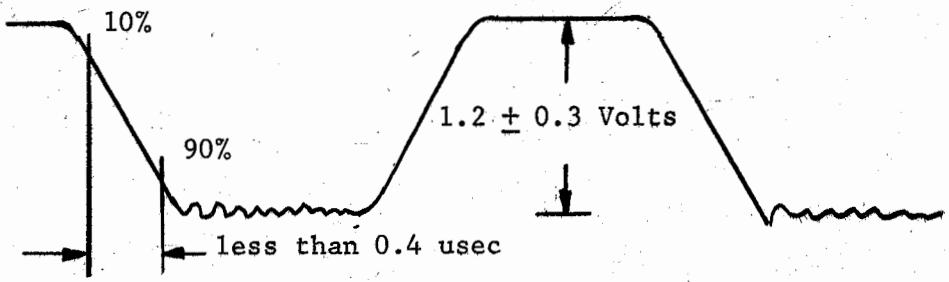
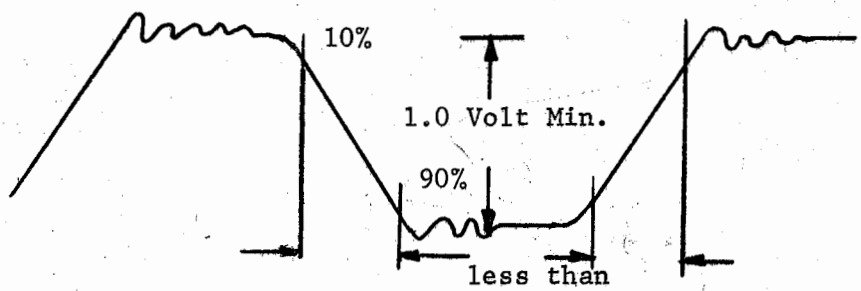


FIGURE 6

7

FIGURE 7



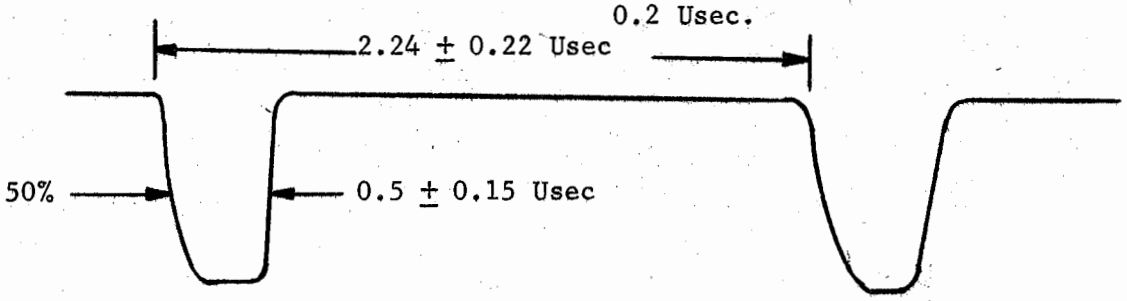


FIGURE 8

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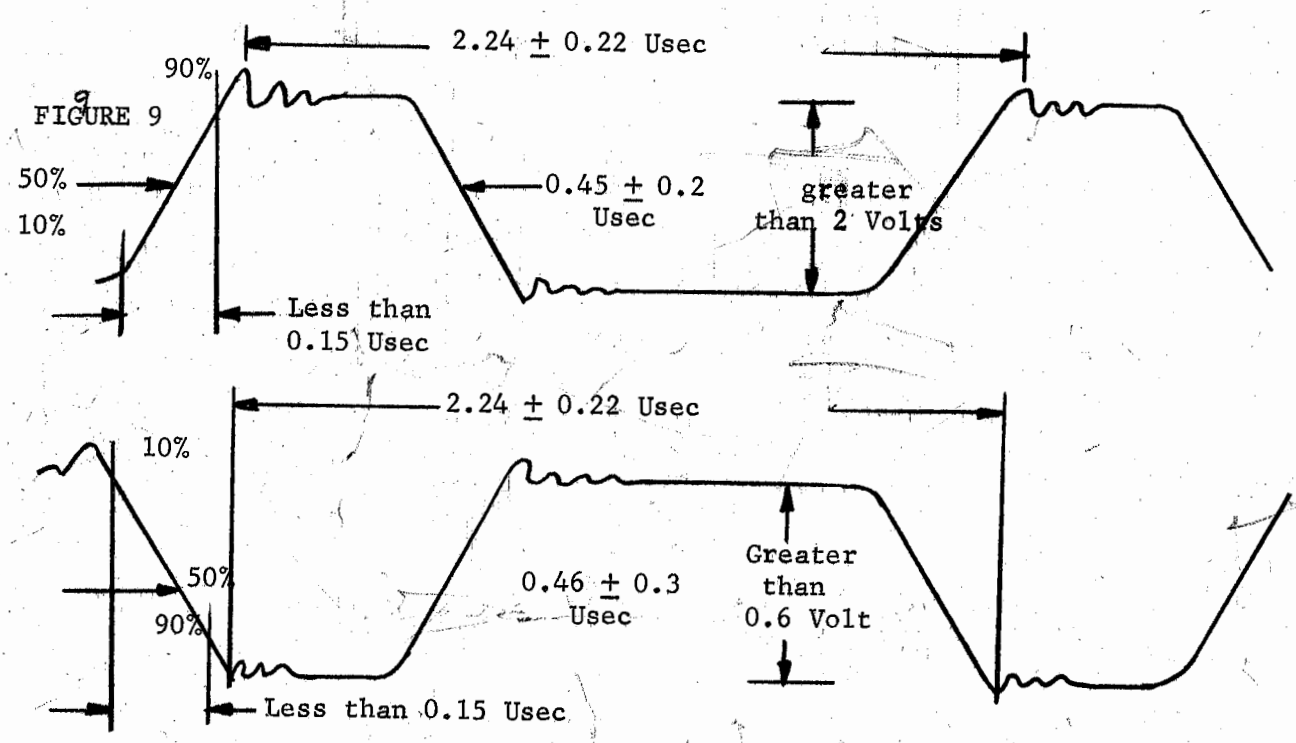


FIGURE 9

FIGURE 10

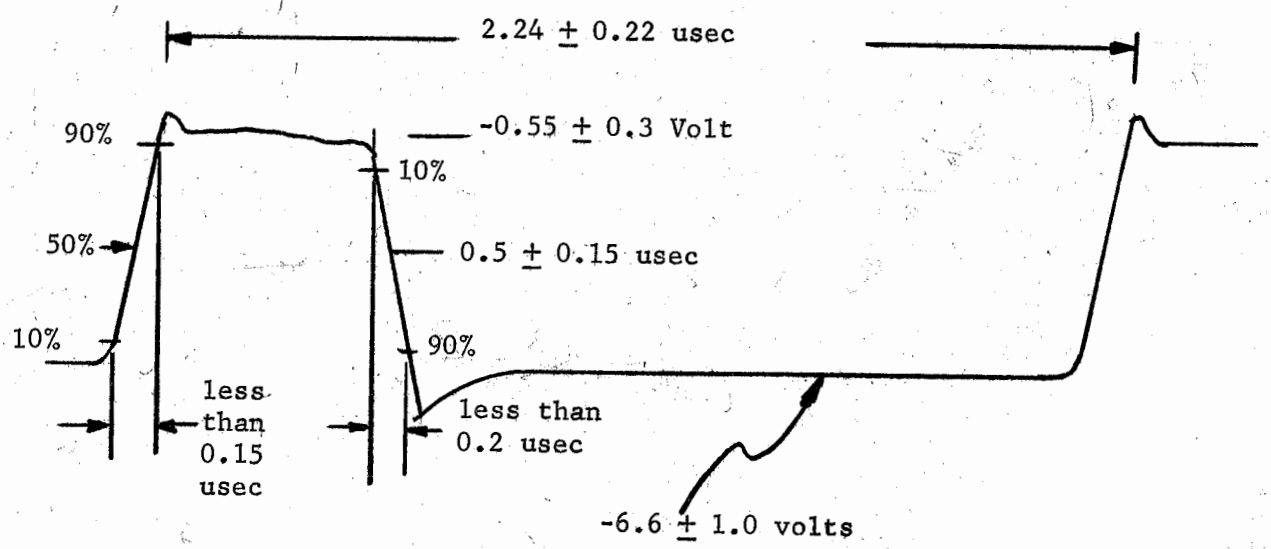


FIGURE 11

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